

An explanation of Swiss Hallmarks on Gold Watchcases

by Louis S. Flocco (PA)

Hallmarks Defined

If a wristwatch can be a status symbol, or an indicator of personal style,¹ then the wearing of a gold watch demonstrates the epitome of status, or perhaps, simply good taste. If you own a gold watch, you may have noticed small divots² on the case back or lugs of your otherwise perfect Swiss watch and wondered what those almost microscopic marks signified; these marks are what we are going to explore.^{3,4}

Since early times, gold and silver (later platinum and then more recently palladium) were used for the manufacture of expensive and high quality jewelry. Similarly, forgeries and imitations of such articles existed from the beginning.⁵ So, in an attempt to protect the integrity of the precious metal objects market, hallmarks were developed. The word hallmark is derived from London's Goldsmiths' Hall of the Worshipful Company of Goldsmiths, the originator of Britain's first hallmarks.⁶ A hallmark is the mark or marks stamped, impressed, or struck on gold, silver, or platinum that indicate its fineness, karat, or purity. Hallmarks have been in use in England since the fourteenth century. While the United States has never used hallmarks, indications of fineness or karat have been required since 1906.⁷ Most European countries use official hallmarks.

In an attempt at standardizing the legislation on the inspection of precious metals and to facilitate international trade, in 1973 and again in 1995,⁸ Denmark, Austria, Finland, Norway, Portugal, Sweden, Great Britain, Ireland, the Czech Republic, the Netherlands, and Switzerland signed the Vienna Convention on the control of the fineness and the hallmarking of precious metal objects.^{9,10} Since 2004, Latvia and Lithuania have joined and Israel, Cyprus, the Ukraine, and Hungary are expected to join in the near future.¹¹ Those articles, which are assayed and found to be in conformity by the qualifying office of a signatory country, receive a mark, known as the Common Control Mark (CCM), attesting to the material's fineness (See Figures 1 and 2). This mark is recognized in all the other contracting States. The multi-tiered motif of the Common Mark is the balance scales, superimposed, for gold, on two intersecting circles, or for platinum, a diamond shape.¹² Each includes a fineness standard.

Figure 1.



Figure 2.



1. This phrase is adapted from the current Seiko ad campaign, "One's watch is like a window into one's personality and, as such, each person decides, through his or her choice of watch, how the world will see them." See Seiko website.

2. See L.F. 941.311 *Ordonnance sur le Contrôle des Métaux Précieux*, hereinafter referred to as ("OCMP") Article 122, if the refinisher polishes out the hallmarks, the process must start all over.

3. See L.F. 941.31 *Loi fédérale* of June 14, 1994, *Loi sur le Control Des Métaux Précieux*, hereinafter referred to as ("LCMP") (effective, July 27, 2004; amended July 27, 2004; original June 20, 1933, superceding *Loi fédérale* December 28, 1880, December 21, 1886, and June 17, 1886). http://www.admin.ch/ch/f/rs/941_31/index.html. See also, L.F. 941.311 *Ordonnance sur le Contrôle des Métaux Précieux*, hereinafter referred to as ("OCMP") http://www.admin.ch/ch/f/rs/941_311/

4. Although beyond the scope of this inquiry, the law on "Swiss Made," *Ordonnance réglant du nom "Suisse" pour les montres* (1995) may be found at L.F. 232.119 http://www.admin.ch/ch/d/sr/c232_119.html; and the law on the Geneva Hallmark, *Règlement concernant l'organisations du bureau de contrôle facultatif des montres de Genève, poinçon de Genève* (1994) may be found at I.1-25.04. http://www.geneve.ch/legislation/rsg/f/s/rsg_I1_25P04.html

5. See *L'Administration fédérale des douanes* (AFD) website, www.zoll.admin.ch.

6. Romero, Christie, *Warman's Jewelry*, 2nd ed., Krause, 1998.

7. See 15 USCS 45 (2005); 15 USCS 291et seq (2005).

8. There were technical amendments made in 1998 as a result of the Houtwipper case.

9. Switzerland still adheres to the terms of the Convention of November 15, 1972, and also has reciprocal pacts with Austria, 1992; Spain, 1935; France, 1987; and Italy, 1970. OCMP Article 126.

10. See AFD website, including illustrations.

11. This information is from personal communications with Daniel Monney of the ADF, hereinafter, "Monney."

12. Although beyond the scope of this inquiry, this is the Common Control Mark for silver



Table 1. Worldwide Gold Fineness Standards

<u>Karats</u>	<u>Fineness</u>	<u>Gold content</u>	<u>Comments</u> ^{21,22}
	<u>‰</u>	<u>%</u>	
24	999	99.9	Canadian Maple Leaf Coins
24	990	99.0	Minimum for Fine (Pure) Gold
22	916	91.6	Standard Indian subcontinent
21.6	900	90.0	Swiss "Vreneli" 20fr coins
21	875	87.5	Standard Arabic countries
19.2	800	80.0	Spanish & Portugese Standard
18	750	75.0	Italian & Swiss Standard
14	585	58.5	U.S. Standard $\pm 3\%$
10	417	41.7	U.S. Minimum
9	375	37.5	U.K. standard
8	333	33.3	German & Austrian Standard

Swiss Hallmark Standards

In Switzerland, the laws on the fineness of precious metals, the Federal Law on Precious Metals Control and its Ordinance,¹³ protect Swiss precious metal watch consumers worldwide by combating violators, forgers, and swindlers. The laws prescribe the criteria for articles of gold, silver, platinum, palladium, multi-metal¹⁴ articles, and plated objects. The law sets forth the minimal and the material requirements and enumerates the penalties for violation of the law.¹⁵ Swiss law makes several discrete provisions for watchcases and their hallmarking. For example, Swiss Law adroitly defines a watchcase as "any object that contains a watch movement."¹⁶ More specifically, the law limits a watchcase to objects housing a timekeeping movement of less than 50mm in diameter and 12mm in thickness (as measured from the bottom of the main plate to the top of the bridges and excluding complications).¹⁷ Otherwise, all articles of gold, silver, platinum, palladium, or alloys of these precious metals must meet a legal standard of fineness and only certain purities are acceptable. In Switzerland, one can make and sell 9, 14, 18, 22, and 24 karat gold jewelry, but not 12 karat gold, as the latter is not a standard recognized by law.¹⁸ In some countries, jewelry lower than 12 karats

(50 percent gold or 500 fineness) cannot be described as gold.¹⁹ Table 1 presents an overview of recognized fineness worldwide.

Swiss Precious Metal Standards

The standard gold fineness in Switzerland is 750Au (with no negative tolerances permitted).²⁰ Thus, most Swiss gold watchcases of modern vintage are wrought of 18 karat gold.^{23,24} This is irrespective of whether fashioned from yellow gold (75% Au + 15% Ag + 10% Cu), rosé/pink gold (75% Au + 12.5% Cu + 12.5% Ag), red gold (75% Au + 20.5% Cu + 4.5% Ag), or white gold.²⁵ Originally developed in the 1920s as a substitute for platinum,²⁶ the content of white gold is not

13. The Federal Laws of the Swiss Confederation are pre-emptive, similar to certain U.S. Federal Laws.

14. "Multi-metal" objects are articles composed of precious metals on a base metal, such as gold-on-steel. Where technically possible, the base metal parts should be labeled, "METAL" or "INOX," etc. ICMP Section 5.3

15. See LCMP Article 17; 44 through 56. Penalties for violations can range from CHF50,000 for negligent violations to CHF100,000 for willful violations. LF 941.31 Chapitre VII.

16. See OCMP Article 38.

17. See L.F. 232.119, "Ordonnance Régissant du Nom 'Suisse' pour les Montres," Article 1-1 (1995). Compare, D.243, L.F. 941.311.1, Instructions on the application of the Precious Metals Legislation (ICMP). Whereas the OCMP Article 38 reserves for the Bureau the discretion to determine what kinds of "movement holders" are watchcases for purposes of the law, the ICMP states in translation that: "a watchcase subject to obligatory official punching within the meaning of article 38 of the OCMP is any object containing a watch movement having a maximum thickness of 12mm and a maximum diameter of 50mm." http://www.admin.ch/ch/f/rs/c941_311_1.html. Anything larger is considered a "clock."

18. See LCMP Annexe 2.

19. See "Caratage of Jewellery." <http://www.gold.org/jewellery/technology/caratage/> (World Gold Council).

20. Monney (in Switzerland 18 karat is 750 fineness, minimum).

21. See, "Caratage of Jewellery," "Some countries insist that there is no negative tolerance allowed (e.g. U.K. where 18 karat is 750 fineness, minimum), but in others a negative tolerance, typically three parts per thousand, is allowed (e.g. in the U.S., a fineness of 747 would be accepted as 18 karat). This causes difficulty in the mutual recognition of national marks/hallmarks, a problem raised in the European Union by the ruling of the European Court of Justice in the case of Ludomira Neeltje v. Barbara Houtwipper (case c.293/93). Thus, a piece of jewelry assaying at 747 fineness would pass in the U.S. as 18 karat, but fail in the U.K."

See also, "Caratage of Jewellery." "At 14 karat, the mark 14KP is found in the USA, the P indicating 'plumb' to differentiate between the U.S. standard and the international standard for 14 karat."

22. Trueb, Lucien, F., "Gold in Watchmaking," http://www.goldbulletin.org/downloads/Trueb_1_33.pdf, page 14.

23. *Swiss Watchmaking Year* reports that in 2002 precious metal watches of which platinum comprised only a few thousand pieces, constituted 5 percent of 26.8 million exported watches, translating to 48 percent of export value. They report in 2003 precious metal pieces numbered 5 percent of 24.6 million exported watches or 46 percent of export value. <http://www.swiss-watchmaking-year.com/statistics/>. See also, Buchanan, *Watch Buying Basics*, Watch Time Buyer's Guide 2005, p. 40.

24. Trueb, p. 13.

25. Monney; Compare "Caratage of Jewellery"; Compare, Trueb, p. 14.

26. See "Jewellery and Technology-Gold Jewellery Alloys," (World Gold Council), <http://www.gold.org/jewellery/technology/alloys/index.html>.

Sidebar 1—Standard Platinum Fineness

The standard fineness of platinum in Switzerland is 950Pt and the standard alloy of platinum for watchcases in Switzerland is (.950Pt + .050Ru or Cu)³³ representing about 90 percent of the production.

legally defined.²⁷ There are two types of white gold alloys, nickel-white (which has fallen out of favor due to the incidence of allergic reaction²⁸) and palladium-white (75% Au + 15% Pd + 10% Ag, Cu, Ni, and/or Zn). While the so-called "white metal" is popular, it is the relatively more expensive palladium²⁹ that leads not only to its appearance as stainless steel, but also creates the disparity in price between gold-colored and white-colored watches. Palladium-zinc-silver alloys are also more difficult to cast³⁰ and are otherwise more difficult to process because of an elevated melting temperature.³¹ It is interesting to note that of all metals, only gold and copper have color, yellow and red respectively. All other metals are gray/white and, in alloy, tend to wash out the gold color, whereas copper imparts reddish hues.³² So it would seem the preference for white metal runs counter-intuitive, as colored metals are the rarities.

In Switzerland, any article in precious metal, a multi-metals article (objects made by a combination of the precious metals and base metals, including aluminum, iron, copper, nickel, zinc, and tin) or plated articles (see Sidebar 2³⁴) that are put into the stream of commerce in Switzerland must carry prescribed designations,³⁵ including a maker's mark, which is properly called a "Responsibility Mark,"³⁶ and a standard of fineness expressed in parts per thousand. The Responsibility Mark (RM) is registered at the Central Office for Precious Metals Control. Only precious metal watchcases must be officially hallmarked. Swiss hall-marking for other articles like jewelry and cutlery is optional. The Official Hallmark for all precious metals and for all the legal fineness standards is the head of a St. Bernard. In addition to the Swiss hallmark, all pre-

Sidebar 2—Swiss Standards for Plated Metals

Plated articles, which were once illegally passed for precious metal articles, are now legally permitted if properly marked and sold for what they really are, and thus not intended to cause confusion. The two general methods of watchcase plating are: galvanic/electrolysis gold deposition and mechanical plating (wherein the gold is physically applied, welded, or rolled onto a base metal). Swiss Law requires plated articles to be marked with a Responsibility Mark (RM), thickness of the plating, and the method of plating. The marks are "L" for laminé for .585 rolled gold plate and "G" for .585 gold plating. The minimum plating is five microns and the actual thickness must be marked on the article, i.e., "Plaqué G 10 mic. + RM" or "P 10 + RM." It is optional to note the metal if not obvious. For watchcases and bracelets, the highest quality plating is at least 200 microns $\pm 20\%$ of .585 gold and marked "C" or "coiffe or," which is generally steel-capped with gold. Markings such as 1/10 R.G.F. or 12k G.F. do not refer to thickness, but rather to the ratio of the weight of the precious metal to the base metal. Such designations are common in the U.S. and Germany, but are not permitted in Switzerland. For plated articles, the application of a standard of fineness is prohibited.³⁷

ciuous metal goods may be stamped with the CCM of the Vienna Convention.

"Hallmarks" vs. "Marks"

The term "hallmark" is often used too loosely. It is important to differentiate between a "mark" and a "Hallmark." They each carry different guarantees. Only an independent third party, typically an Assay Office, applies a hallmark after the item has been assayed for gold content and found to be within the legal tolerance.³⁸ Swiss Law does not permit negative

²⁷ See, generally, Thompson, Michael, "Defining White Gold," (*Professional Jeweler*, September 2003) www.gold.org/jewellery/technology/pdf/09.Metalsmithing.sa.pdf

²⁸ See Raykhtsaum, G. & Agarwal, D. P., "Nickle Release Tests-How Well do they Work?," a paper presented May 2001 at the Santa Fe Symposium.

²⁹ See Trueb, p. 14, Palladium varies in price due to cyclical uncertainties of the Russian supply, which provides 70 percent of the world's palladium and the increased demand for it for automotive catalytic converters.

³⁰ See Corti, as cited in by Thompson, "Defining White Gold."

³¹ See "Gold Jewellery Alloys," table 7.

³² See "Gold Jewellery Alloys."

³³ Monney. Compare, "Platinum Jewelry Alloys," *Platinum* 2002, pp. 28, 29.

³⁴ See LCMP Annexe 1-1; OCMP Article 43-2, 3, 4; Article 49. See also, AFD website.

³⁵ See LCMP, Articles 7 through 12.

³⁶ The official German term is *Verantwortlichkeitsmarke*, literally translating as "responsibility mark."

³⁷ Monney.

³⁸ See OCMP Article 117-1, 2, and 3.

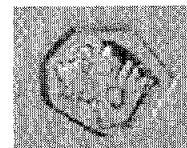
tolerances. In the first instance, however, the manufacturer must stamp each article with a standard of fineness and its Responsibility Mark.³⁹ After assaying, an office of the Precious Metals Control can strike the Swiss Hallmark. Thus, indications of the fineness and the Responsibility Marks are “marks,” whereas the St. Bernard and the Common Control Mark are “hallmarks.” Notwithstanding that the location of some “marks” may be inside the watchcase, the control office applies the official hallmark next to the responsibility and the fineness marks. One official hallmark per article is sufficient if visible on the exterior of the product. For watchcases where the manufacturer marks on the inside, an additional fineness mark must appear on the exterior. In this case, the Control Office strikes the official mark on both the inside and outside parts of the watchcase. The additional fineness indication on the exterior of the watchcase is not necessary if the Common Control Mark is to be applied, since this includes a fineness standard.⁴⁰ The French term “poinçon” means “punch,” but loosely translates to “hallmark.” In French it refers both to the steel awl or punch used to make the mark, decoration, or design in metal, as well as the mark itself on the article.

The official hallmark, *poinçon officiel* is struck by certified-assayers⁴¹ on the articles whose fineness is found in conformity to the law.⁴² In other words, the official hallmark is the official guarantee of the fineness by the State.⁴³ The official Swiss hallmark is the head of a Saint Bernard.⁴⁴ The dies for it are made at the Central Office for Precious Metals Control in Bern under strict security.⁴⁵ On the lower rear side of the dog's ear is a place for the specific mark of the branch assayer's office,⁴⁶ which is similar in concept to a mint-mark on coins. There are currently 11 assayer's offices

Figure 3A.



Figure 3B.



located in the Swiss cities of: Berne (B), Basel/Bâle (), Biel/Bienne (B), Chiasso (T), Genève Montbrillant and Genève Aéroport (G), La Chaux-de-Fonds (C), Le Noirmont (J), Schaffhausen (S), Zürich and Zürich-Flughafen (Z).⁴⁷ The sign in parenthesis following the city is the special mark indicating where the object was assayed⁴⁸ and it is punched in place of the “x” on the dog's head, as shown in Figure 3A. In Figure 3B, from an 18k Omega, the “J” for Le Noirmont is almost visible. In the past⁴⁹ these special marks were also used: Central Bureau at Berne (✚), Basel Aéroport (M), Granges/Grenchen (g), Chiasso (t), Genève (G), Genève Aéroport (A), Lausanne (V), Le Noirmont (n), Le Locle (L), Neuchâtel (N), Romanshorn (R), and Zürich-Flughafen (K) and (Z). See Table 2. The majority of the branch assayer's offices, current and historic, with the exception of Chiasso, Romanshorn, and Zürich tend to be located on the western border of Switzerland, from Genève in the south to Schaffhausen in the north, along what is referred to as *l'Arc jurassien*, the historic watchmaking arc of the Jura Mountains.

Earlier Swiss Hallmark Symbols

Certain historic assayer's offices are no longer maintained by *Bureau Central du Contrôle des Métaux Précieux*,⁵⁰ but are nonetheless legitimate for watchcases hallmarked in the past.⁵¹ Similarly, before the adoption of the Saint Bernard, there was an old “set” of

39. See LCMP Article 16.

40. See OCMP Article 117-1, 2, and 3. *Un poinçon officiel au moins doit être visible à l'extérieur des boîtes de montre poinçonnées. Les boîtes de montre sur lesquelles le fabricant a apposé le marquage complet à l'intérieur doivent porter une indication de titre supplémentaire à l'extérieur. Dans ce cas, le bureau de contrôle insculpe le poinçon officiel “tête de saint-bernard” à l'intérieur et à l'extérieur de la boîte. Un seul poinçon officiel extérieur suffit. L'indication de titre supplémentaire à l'extérieur de la boîte n'est pas nécessaire si le poinçon commun de la Convention de Vienne est demandé, car celui-ci contient déjà l'indication du titre.*

41. In French the title is *Essayeur-juré* and is sometimes translated in the British dialect as “Sworn-in assayer.” See, OMCP Article 8 and Article 87. See AFD website for illustrations. Photo of hallmark on Omega courtesy of Ed Hahn.

42. See LCMP Articles 14, 15.

43. See LCMP, June 20, 1933, Articles 13-2; 15-1 (superceded) wherein the poinçon official is parenthetically referred to as the *poinçonde garantie*.

44. The Official Hallmark has two sizes 1.6mm x 2mm or 0.8mm x 1mm. See LCMP Annexe 2, Figure 1 and OCMP Article 58.

45. See LCMP Annex 2, Figure 2. See OCMP Articles 4k; 17; 81; 82.

46. See LCMP Article 15-2.

47. See ADF website. See also, OCMP Article 109 and Annexe 2, figure 3.

48. See OCMP Article 109.

49. Monney.

50. See OCMP Article 3, the Assayer's Office is under the direction of the Customs Bureau.

51. Mooney, see also, the English Hallmarking Act 1973 (prior version Vienna Convention) at Order 1976 recognizing old official Swiss hallmarks at Schedule 1. See AFD website for illustration. It is clear that these hallmarks were used between 1934 and 1995. There is empirical evidence that they were used substantially prior to that time. There is no doubt that the first Swiss hallmarking laws came into effect in 1881 and were amended in 1886 and 1933.

* = Only used for watchcases

Gold 750 Helvetia	Gold 585 Squirrel	Silver 925 Duck	Silver 800 Wood Grouse	Platinum 950 Ibex	Gold 375* Morning Star	Gold 750 Lynx 1	Gold 585 Lynx 2	Silver 925 Gentian 1	Silver 800 Gentian 2	Platinum 950 Rabbit's Head	Gold 375* Titre bas	Gold for export The Lady from Bern	Silver for export The Bear

Figure 4. Official Swiss hallmarks before August 1, 1995.

Swiss hallmarks. The old hallmarks were a set of more concise pictograms, each singularly providing the national identity, fineness, and in which market the article was manufactured. On each was an area reserved for the branch assayer's mark. Figure 4 illustrates the official Swiss hallmarks used before the revision of the law in 1995,⁵² namely from left to right: Helvetia (750Au), Squirrel (585Au), Canard/Duck (925Ag), Wood Grouse, (800Ag), Ibex (950Pt), Morning Star (375Au), Lynx1 (750Au, imported), Lynx 2 (585Au, imported), Gentian 1 (925Ag, imported), Gentian 2 (800Ag imported), Rabbit (950Pt, imported), Titre Bas (375Au, imported), the Lady from Bern (a gold fineness not accepted in Switzerland but intended for export), and the Bear (a silver fineness not accepted in Switzerland but intended for export). Although the current system requires only the official hallmark, nearly all watchcases are also marked with the optional common control mark.⁵³ The current system is a simplified version, which makes no distinction between articles intended for export, import, or domestic use, as the system is international.⁵⁴

Although the table of hallmarks at right is fairly complete, dating as far back as 1973, and for some period prior thereto, the branch assayer's mark on a particular watchcase may not necessarily correspond to the city or region where the watchmaker is located. This is primarily due to the fact that the mark corresponds to the assayer's office where the watchcase was tested; watchcases are hallmarked at the office in or near the place of manufacture.⁵⁵ In the example illustrated, Omega is located in Biel/Bienne, but further inspection would reveal that the watchcase was made and, therefore, hallmarked at Le Noirmont. Omega does not manufacture its own watchcases, but rather buys them from a watchcase manufacturer, even if that watchcase manufacturer, just as its ébauche manufacturer, is a subsidiary of its parent Swatch Group. Only producers of *Haute Horlogerie* such as Patek Philippe,

Table 2. Assayer's Office Mark's Stamped Inside Hallmarks

Mark	Assayer's Office	Dates of use
★	Basel/Bâle	Pre 1973 to Date
✚	Berne	Pre 1973-1994
A	Genève Aéroport	Pre 1973-1994
B	Berne	After 1995
B	Biel/Bienne	Pre 1973 to Date
C	La Chaux-de-Fonds	Pre 1973 to Date
G	Genève Montbrillant	Pre 1973 to Date
G	Genève Aéroport	After 1995
g	Granges/Grenchen	Pre 1973-1994
J	Le Noirmont	After 1995
K	Zürich-Flughafen	Pre 1973
L	Le Locle	Pre 1973 to Dec. 2001
M	Basel Aéroport	Pre 1973-1994
N	Le Noirmont	Pre 1973-1994
n	Neuchâtel	Pre 1973 to Dec. 2001
R	Romanshorn	Pre 1973 to May 1997
S	Schaffhausen	Pre 1973 to Date
T	Chiasso	After 1995
t	Chiasso	Pre 1973-1994
V	Lausanne	Pre 1973 to Dec. 1995
Z	Zürich	Pre 1973 to Date
Z	Zürich-Flughafen	After 1973 to 1994
Z	Zürich-Flughafen	After 1995

Grey highlighted entries are for current Assayer's Offices.

Piaget, Rolex, and some others make their own watchcases. A few small manufacturers also fabricate their own cases. In particular, Parmigiani of Fleurier has artisans in La Chaux-de-Fonds who have, at one time or another, created complete, polished, decorated, and crowned cases for Bréguet, Girard Perregaux, Parmigiani, Ulysse-Nardin, and Vacheron Constantin.⁵⁶

52. Vienna Convention, June 19, 1994. The old marks were used prior August 1, 1995.

53. Monney.

54. See ICMP, the international marks save time, reduce cost, simplify import procedures, and carry international guarantees. See also, LCMP, June 30, 1933, Articles 4, 15, and 19 (superseded).

55. Monney.

56. See Trueb, pp. 17,18.

Case Manufacturers Operate Behind the Scene

The norm is that inasmuch as many Swiss watch companies do not produce their own ébauches, similarly they do not refine their own precious metal watchcase blanks.⁵⁷ Indeed, it is considered a great esteem to have in-house case fabrication facilities. Still, Swiss watchcase manufacturers buy most of their raw gold parts from the major suppliers: Precinox in La Chaux-de-Fonds, Metalor in Neuchâtel, Valcambi in Balerna, Argor-Heraeus in Mendrisio, Varinor in Delémont, and Cendres & Métaux, in Biel/Bienne.⁵⁸ Even Patek Philippe, universally acknowledged as a most complete in-house operation, has been known to purchase pre-shaped case rings with lugs and perhaps case-backs, bezels, and bracelet links from specialized suppliers, which are then cold-worked, milled, cut, and polished in-house.⁵⁹ Since 2001, Patek Philippe has owned its own watchcase company, Calame & Cie., in La Chaux-de-Fonds. Similarly, the Swatch Group controls two gold watchcase and bracelet fabrication facilities: Favre-Perret in Le Crêt du Locle (founded in La Chaux-de-Fonds in 1865 and acquired in 1999)⁶⁰ and Lascor SpA, in Sesto Calende in Northern Italy (specializing in cases and bracelet production in steel, aluminum, and precious metal composites).⁶¹ Lascor's machining processes include case stamping, forging, machining, burnishing, and polishing.^{62,63} Lascor's cases are imported into Switzerland from Italy at Chiasso, but must comply with the Swiss Laws and be assayed and hallmarked⁶⁴ at the Precious Metals Control Office in Chiasso.⁶⁵ Both Swatch Group divisions also produce cases for other watchmakers not related to the Swatch Group.

Although there has been significant vertical consolidation of the watchcase sector into the watchmaking industry in the last handful of years, even the fully integrated watchmaking giants cannot always supply themselves through their own subsidiaries and must look elsewhere to meet their production needs.⁶⁶ For the most part, about 10 or 15⁶⁷ niche companies out of some 50 specialized case makers supply about 90 percent of the watchcases used in Swiss production.⁶⁸ Many of those watchcase manufacturers belong to the trade group, *l'Union Suisse pour l'Habillage de la Montre* (USH). Table 3 shows an overview of some gold watchcase makers along with their affiliates, but not necessarily their customers.⁶⁹ For as beautiful and complicated as the watchcases are, watches carry the name of the watchmaker; the case maker is not overtly identified.⁷⁰ Watchcase manufacturers choose to live in the shadow of the watch brands, not daring to acknowledge the names of their prestigious customers.⁷¹

The Casemaking Process

The general process employed by most case fabricators involves taking gold ingots and rolling them down in size, annealing, and then re-rolling to the required thickness. The fabricators produce flat and ring-shaped parts in the vague shape of the watchcase to be. The cold-working and forging of precious metals is an extremely demanding job that is undertaken only by highly specialized companies.⁷² Surprisingly, gold is perhaps not as "soft" as popularly believed since, for example, in the cold-worked state 18k gold can achieve a Vickers' Hardness rating in the range of 214 to 227,⁷⁴ as compared to the more heavily alloyed 9k, which falls into the range 135 to 165. To clarify, in contrast to watchcases, most gold jewelry is cast,⁷⁵ which may par-

57. The law at LCMP, Article 9-2, recognizes this fact. Accord, 1933 version.

58. Monney.

59. Trueb, pp. 16, 17.

60. See The Swatch Group, Ltd website, <http://www.swatchgroup.com/companies/companies.php>.

61. See Swatch Group. Ruedin was acquired in 1998 and makes steel cases, but it made gold cases before the acquisition.

62. Trueb, pp. 13, 16.

63. See Swatch Group.

64. Trueb, pp. 13, 16. See LCMP Article 20. See also OMCP Article 126-2.

65. Monney.

66. See Kuenzi, Françoise, "Habillage, La Boîte Tient la Forme," (*l'Express* 2004). <http://www.lexpress.ch/services/horlogerie/2004/mondehorlogerie/habillagelaboite.htm>

67. See Donzé, Gerald, "Parole d'Experts: Boîtes de Montre, (Watches-Lexic)"; <http://www.watches-lexic.ch/pages/fr/tec/exp9.htm> Compare, Trueb, p. 14 (indicating 15 gold watchcase makers).

68. See Kuenzi.

69. See Droz, Daniel, "Maîtres-horlogers dans la Jura," (*l'Express* June 2003). <http://www.lexpress.ch/services/horlogerie/2003/juin/lacroix.htm>

70. See Donzé. Compare, OMCP Article 75.

71. See Kuenzi.

72. The identification number and affiliates information courtesy of Monney. For employee information, see OSEC—the Swiss Export Directory. www.osec.ch.

73. Trueb, p. 15.

74. See Gold Jewellery Alloys.

75. See Süss, Linger, Glaner and du Toit, "18 Carat Gold Alloys with Increased Hardness," Gold Bulletin 2004, 37/3-4 (World Gold Council). http://www.goldbulletin.org/downloads/Suss_3_37.pdf.

Table 3. The Major Case Makers

Case maker	City/Region	Affiliate/note ⁷²	Identifier
Adea Fontan	Les Acacias	® 1934	5/27
AS Line	St-Imier	® 1991	1/370
BGM Miserez	Saignelégier	® 1934	1/163
Bruno Affolter	La Chaux-de-Fonds	Sandoz (Parmigiani) (2004); ® 1986	1/361
Calame & Cie.	La Chaux-de-Fonds	Patek Philippe (2001) ® 1979	1/357
Cartier	La Chaux-de-Fonds	Richemont Group (closed) ® 1934 ® 1934 ® 1934	1/104 1/124 1/221
Cortech	Cornol	LVMH Group using customer's RM	RM
Donzé- Baume	Les Breuleux	101-150 workers; ® 1987	1/363
Ecodor	Plan-les-Ouates	Frank Muller; ® 1991	5/45
EMG Micromécanique	La Chaux-de-Fonds	Girard-Perregaux; ® 1990	1/366
F de G.	Chêne-Bourg	® 1983 ® 1983	1/359 5/42
Fabior, Oliver Judas	Gimel	® 1957	5/7
Favre-Perret	Le Crêt du Locle	Swatch Group (1999); ® 1934	1/115
Florian Favre	Les Acacias	10-30 workers; ® 1989	5/43
Froidevaux frères	Müntschemier	® 1992	1/371
G. Meuwly & Fils	Genève	® 1934	5/34
Granjean Sarl	La Chaux-de-Fonds	® 1973	1/354
Guillod Gunther, SA	La Chaux-de-Fonds	Bréguet; 51-100 workers; ® 1934 ® 1934	1/117 1/121
Guyot & Cie	La Chaux-de-Fonds	® 1934 sold 2003	1/122
H. Heggendorff	Grenchen/Granges	® 1934	1/151
Jaquet	La Chaux-de-Fonds	® 1985	1/360
Jean Arnoux	Le Noirmont	® 1934	1/158
Jean-Pierre Hadorn	Bile/Bienne	® 1994	1/374
Joseph Erard	Le Noirmont	31-50 workers; ® 1934	1/161
JSG SA	La Chaux-de-Fonds	® 1934	1/129
Louis Lang	Porrentruy	Independent; 401-500 workers; ® 1989	1/365
M. Neuenschwander	Le Locle	® 1994	1/376
Manufacture Ruedin	Bassecourt	Swatch Group; 151-200 wrkrs; ® 1987	1/364
Miguel Gil	La Chaux-de-Fonds	® 1993	1/372
M.P.R., S.A.	Alle	® 1993	1/373
Nouvelle Piquerez	Bassecourt	151-200 workers; ® 1963	1/183
Orac (A. Bräuchi)	Yvonand	® 1934	1/126
Oréade	La Chaux-de-Fonds	high-quality watchcases; ® 1934 ® 2003	1/109 1/122
Patek Philippe	Genève	With 401-500 workers; ® 1934	5/28
Paul Wyss	Courtételle	® 1999	1/377
Piaget, SA	Plan-les-Ouates	Piaget (Richemont), formerly Prodor	RM
Pic	Le Noirmont	® 1991	1/369
Ponti & Cie	Genève	11-30 workers; Richemont; ® 1934	5/21
Queloz SA	Saignelégier	Maurice Lacroix (1989); ® 2003 ® 1962 through 66	1/378 1/177
Revor, W. Brusa	La Chaux-de-Fonds	® 1934	1/351
Rolex, SA	Genève	Private production uses own RM	RM
Rubattel & Weyermann	La Chaux-de-Fonds	® 1934	1/210
Simon & Membrez	Courtételle	® 1978	1/356
Stila	La Chaux-de-Fonds	With 31-50 workers; ® 1934	1/353
USH	Biel/Bienne	Trade group; ® 1969 ® 1934	1/100 5/10
Valentini SA	Bassecourt	® 1990	1/367
Varin étampage	Delémont	Large Production; uses customer's RM	RM
Weber & Cie.	Les Acacias	® 1934 ® 1958	5/29 6/29

tially explain why the Swiss laws make discrete provisions for watchcases. The machining of precious metals is done with dedicated machinery.⁷⁶ The forging tools are prone to wear as the processes involve repeated stampings at pressures in the range of 80 to 120 tons.⁷⁷ This is not too dissimilar from the coin minter's process, but is much more involved⁷⁸; a solid gold watch can include a gold watchcase, case-back, bezel, dial, hands, crown, pushers, bracelet, and/or clasp.⁷⁹

The Central Bureau permits the optional marking of precious metal watch parts such as dials, movement parts (for example rotors), as well as crowns or push-pieces, with the fineness and responsibility marks. The Bureau also adheres to the "fastening principle" for watchcases. On a watchcase made entirely of precious metal, the different constituent parts must be in direct contact with each other, for example gold back on gold middle. However, base metal parts, such as movements and movement parts, dials, crowns, winding-shafts and push-pieces, bars for fixing the bracelets to the wrist-watches, other spring parts, screws for fixing the watch back, screws for changing the length of the bracelets, separable tubes for winding-mechanisms on watchcases made of gold, platinum, and palladium, movement holders, and casing rings may be used. Similarly, white gold parts are permitted on platinum watchcases and bracelets, including white gold tubes for winding-mechanisms fixed in an inseparable manner (soldered or pressed in). The Central Office may allow other white gold parts on watchcases and bracelets made of platinum when their function is to fasten, secure, or when they are used as springs, if a case is submitted for consideration with accompanying samples or technical drawings.⁸⁰

The specialized case manufacturers must machine the intricate inner recesses of the cases to accept the bezel, crystal, crown, movement, and case-back, which involves milling, drilling, lathing, and sometimes the tapping of the case-back threads.⁸¹ Polishing, however, is done by hand, as there is no economy of scale for

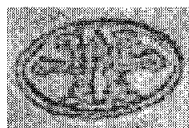


Figure 5A.



Figure 5B.

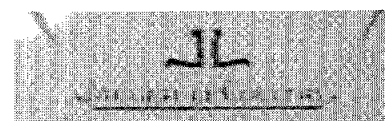


Figure 5C.



Figure 5E.



Figure 5D.

Figures 5A-5E. A trademark and Responsibility Marks.

machine operations for what are usually small series items.⁸² The case manufacturer completely assembles the watchcase, including the crystal, and delivers it to the watchmaker for casing-up.⁸³

The Watch Case Responsibility Mark

Notwithstanding that most watch companies do not make their own watchcases, all precious metal watchcases must nonetheless carry a Responsibility Mark.⁸⁴ The Responsibility Mark or *poinçon de maître* (Figure 5A) is conceptually similar to a traditional trademark (Figure 5B) and by definition should be easily identifiable. The Responsibility Mark (RM) as well as the trademark can be equated to a signature, however, a distinction must be drawn between the trademark⁸⁵ and the Responsibility Mark. The mark illustrated in Figure 5A is the Responsibility Mark from inside the dust cover of a Jaeger LeCoultre white gold watchcase. The logo is a stylized "JL" and a pine tree inside an oval, but it relates only to precious metal watchcases. When watch aficionados describe a watch as being "triple signed," they are referring to the appearance of the maker's trademark (name or stylized logo) that appears on the dial (Figure 5B), crown (Figure 5E), and

76. Trueb, p. 16.

77. Trueb, pp. 14, 15, 16.

78. Harper, ed. *North American Coins & Prices 9th Ed.*, Krause 1999, p. 24 (a U.S. cent requires a striking pressure of 30 tons/in. and some U.S. silver dollars required a striking pressure of 150 tons/in). The Discovery Channel show, *Making Money*, toured the Royal Canadian Mint in Ottawa where Maple Leaf gold coins are struck from gold, and refined to a fineness of .995 to .99999. The gold is cast into 50-pound ingots, which are rolled more than a dozen times until they reach the thickness of the coins to be struck. The presses that make one-ounce Maple Leaf gold coins strike them with 133 tons of pressure.

79. Trueb, p. 14.

80. See ICMP, Supplemental Provisions for Horology, Section 4.3 et seq.

81. Trueb, p. 16.

82. Trueb, p. 16.

83. Trueb, pp. 16, 18.

84. See LCMP Articles 9-2; 9-3; OCMP Article 82. Photo of JL Responsibility Mark courtesy of Richard Madison of Time Zone. Photos of the JLC movement and crown courtesy of John Davis of www.thePuristS.com.

85. Although beyond the scope of this inquiry, the Swiss Law regarding the intellectual property of trademarks can be found at LPM Loi fédérale 232.11.

clasp.⁸⁶ There may also be a visible trademark on the movement (Figure 5D), especially since the display-back is in vogue (and the marks on the movement are required by other laws). Precious metal watches have yet an additional signature, the Responsibility Mark, ostensibly bringing the count to five for a watch with an in-house movement. By affixing a Responsibility Mark to a case, in a very real and legal sense, the marker becomes the guarantor of the accuracy of its metallurgical composition and takes the responsibility for it, including the consequences of any future dispute. The Responsibility Mark constitutes the basis of legal action in the event of violations of the law.⁸⁷ This may partially explain the vigor with which many brands, in conjunction with the Assayers Office, Customs Office, and the Federation of the Swiss Watch Industry, defend themselves against “replicas.”⁸⁸ Manufacturers or *établisseurs* who do not make precious metal watchcases themselves must affix some traceable Responsibility Mark on the article later.

Watchcases prepared by the many specialized case makers of the *l'Union Suisse pour l'Habillage de la Montre* (USH) use one of the six illustrated registered “responsibility marks”⁸⁹ with a “distinguishing number”⁹⁰ inscribed (see Table 3), similar to the local assayer’s special mark. This is called the *poinçon de maître collectif* or collective responsibility mark and, like the RM, guarantees the origin of the product.⁹¹ This mark may be punched on the inside or the outside of the watchcase, but always near the standard of fineness,⁹² which is expressed in parts per thousand in Arabic numbers.⁹³ At the assayer’s office the movement is not cased-up and markings inside the watchcase are visible.⁹⁴ Most makers, however, mark on the exterior.⁹⁵ All

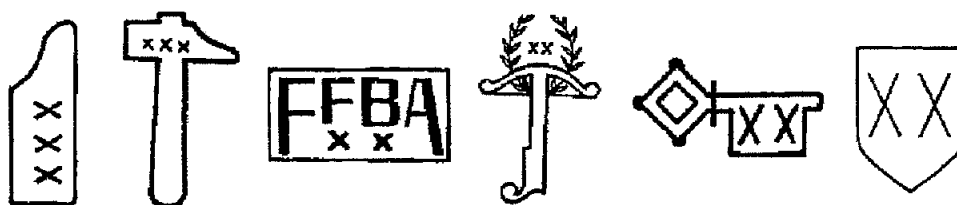


Figure 6. USH Case makers “collective Responsibility Marks.”

of the official and non-official markings are required to be in close proximity to one another. The official hallmark and a fineness standard must be visible to the consumer. However, the maker is permitted to mark on the interior of the watch-

case. If the maker stamps its Responsibility Mark and its claim of fineness on the interior of the watchcase, the maker is further obligated to have a fineness standard visible on the exterior. This obligation may be discharged by requesting the assayer’s office to strike the optional Common Control Mark of the Vienna Convention, with its self-contained fineness standard, on the exterior of the watchcase.⁹⁶ In this situation, the official hallmarks and a standard of fineness will appear twice. Thus, if on a first inspection only the St. Bernard and the CCM are found on the exterior of the watchcase, this would be appropriate; the Responsibility Marks will be inside the case back.^{97,98}

Because the law requires a precious metal watchcase to be traceable to its fabricator, the *poinçons de maître collectif* are public information. The first symbol

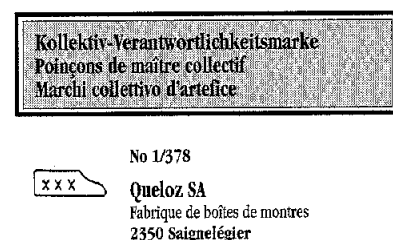


Figure 7. Excerpt from case makers table. See Appendix 1.
Fuille officielle suisse du commerce.

86. The familiar Jaeger-LeCoultre signature is trademark No. 467,503, registered December 6, 1999, and published in the FOSC issue 256.

87. See LCMP Articles 10(b), 44.

88. *Swiss Watchmaking Year 2004* reports that imitations account for a total loss in earnings of 800 million Swiss francs per year. Accord, Monney.

89. Compare, LPM Article 22, and LCMP Article 3 and 9 through 12. Although both laws refer to a “collective mark” they cannot be read *in pari materia*, as the *poinçon de maître collectif* is specific to precious metal watches.

90. See LCMP Article 9.1 through 9.3. OMCP Article 58 through 75

91. See LCMP Article 9 through 12. See AFD website, for illustrations.

92. See OCMP Article 117.

93. See OCMP Article 46. Karatage (fineness expressed in karats i.e. “18k”) is optional since it was long a custom in the industry.

See also OCMP Article 130.

94. See OMCP Article 126.

95. Monney.

96. See OCMP Articles 46-1, 2; 117.

97. See, OCMP Article 117-1, 2 and 3.

98. Any reference or model numbers on precious metal watchcases must be clear enough not to be confused with the standards of fineness, karats, thickness, or microns. ICMP, Article 6.5.

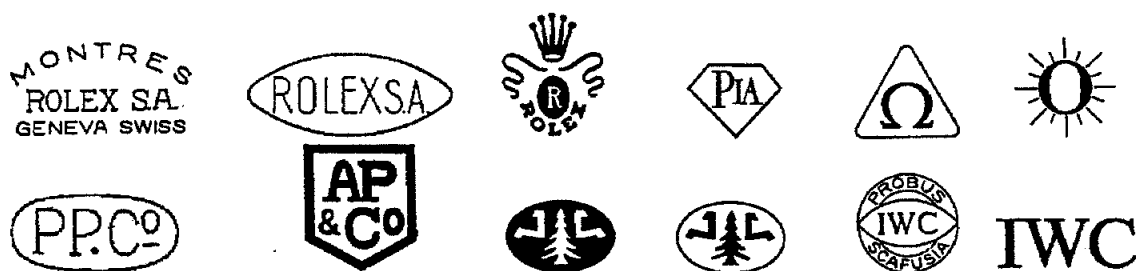


Figure 8. "Individual Responsibility Marks."

of the six shown in Figure 6 is a hammer-head and, as published in the official legal journal, *Fuille officielle suisse du commerce*, see Appendix I, is used by the watchcase maker, Queloz of Saignelégier, a division of Maurice Lacroix since 1989.⁹⁹ Queloz uses the identifier "378." This number inside the hammer-head would reveal that Queloz supplied a particular watchcase. The ubiquitous hammer-head symbol is actually "Collective Responsibility Mark Number 1." This mark has been widely used by many watchcase makers since 1934. It further signifies that the watchcase is made of gold or platinum with a thickness of at least .15mm, and was produced in either the Jura or Neuchâtel Region. The individual case makers are distinguished within the trade group by the two- or three-digit identifier (see Figure 7; Table 3). Of the marks, from left to right: No.1 Hammer-head; No. 2 Hammer; No. 3 Marquee; No. 4 Crossbow; No. 5 Key; No. 6 Shield, only the Hammer-head, Key, and Shield are used for gold watchcases. The Key and the Shield denote production in Genève. The hammer has not been used since 1966.¹⁰⁰ See endnote for greater details.

Keep in mind that case makers not a part of the USH, for example and most notably Piaget, Patek Philippe, Audemars Piguet, and Rolex, use individual responsibility marks and, if they did not make the

watchcase, there should be a collective mark as well. A very small sampling of the many, perhaps familiar, Individual Responsibility Marks are illustrated in Figure 8.¹⁰¹

The logos in Figure 8 represent Individual Marks, from left to right: Rolex mark Nos. 1,404, 644 and 6,468; Piaget mark No. 20,229; Omega mark No. 2,658; Patek Philippe mark Nos. 1,664 and 3,181; Audemars Piguet mark No. 21,447; Jaeger LeCoultre mark Nos. 3,748 and 3,997; and International Watch Company mark Nos. 21,144 and 21,145.¹⁰² While there may be some noticeable similarities in motif, theme, or design, these precious metal watchcase marks are distinguishable from the watchmakers' trademarks, yet in most instances, still instantly recognizable as identifying the watchmaker. For comparison, illustrated in Figures 9 and 10 are the Registered Trademarks of IWC: Trademarks No. 390,470 and No. 390,469, respectively. The Individual Marks are, however, reserved for precious metal watchcases only. Bracelets will have other marks.¹⁰³ For example, in Figure 11 Rolex uses Mark No. 21,424 for its bracelets. Moreover, the collective responsibility marks are not allowed on bracelets. This selection of marks is purely illustrative and is not nearly exhaustive, even for those companies noted.

Assuring Veracity of Marks

Each Assayer's Office has an electronic file of all the recorded trademarks, including the collective makers' marks. There are more than 10,000 Swiss and foreign hallmarks in the registry. This registry is public, at least in the sense that anyone can get information about a particular mark, if one knows what to look for.¹⁰⁴ The marks are also published



Figure 9.



Figure 10.

Registered trademarks for IWC.



Figure 11.

A Rolex bracelet mark.

99. See Droz. See, FOSC, No. 196, October 13, 2003, for illustration.

100. The mark also tells something about the construction of the watchcase. No. 1 Hammer-head is for a gold case of at least 0.15mm thickness and made in the Neuchâtel and Jura Regions; No. 2 Hammer is for cases from Le Noirmont or La Chaux-de-Fonds, but has not been used since 1966; No. 3 Marquee of Fédération des Fabricants de Boîtes Argent (FFBA), for silver watchcases from Neuchâtel and Jura; No. 4 Crossbow, for silver watchcases from Neuchâtel and Jura; No. 5 Key is for gold or platinum watchcases of at least .3mm in thickness from Genève; and No. 6 Shield is for gold cases of less than 0.3mm thickness from Genève.

101. Courtesy of Monney.

102. See LCMP Article 10; ICMP section 3.2.

103. Individual Responsibility Marks do in fact appear on bracelets and deployant clasps and even on some movements. This would appear to be a technical violation of the law.

104. See LCMP Article 12(d) and OMCP Article 75. See AFD website.

Table 4. Watchcase Marking Requirements

Status of Mark	Mark	Struck by	Position	Required
Poinçons officiels Official Hallmarks	Common Control Mark 	Assay Office	External	Optional
	St. Bernard Dog 	Assay Office	External	Yes
Poinçons Marks	Branch Assay Mark	Assay Office	External	Included above
	Fineness, ‰ 750	Maker	Int./Ext.	Yes
	Individual Responsibility Mark 	Maker	Int./Ext.	Yes (Alternate)
	Responsibility Mark 	Maker	Int./Ext.	Yes (Alternate)
	Karatage (k)	Maker	Int./Ext.	Optional

for the purpose of legal notice, in the official daily legal loose-leaf serial, *Feuille Officielle Suisse du Commerce*.¹⁰⁵

The buyer of a Swiss gold watch has redundant assurances that the watchcase was inspected and checked at various levels (by the case makers, watchmakers and certified assayers) and marked with not only meaningful assurances but also the means to trace the origin of the watchcase. Unfortunately, underkarating of jewelry is not uncommon in some countries, including the U.S.,¹⁰⁶ where the marking of jewelry is legally permitted to be done by the manufacturer without any independent checks. In those situations, conformity of fineness is not guaranteed; the consumer relies solely on the jeweler's integrity.

Each gold, silver, platinum, or palladium watchcase introduced into the stream of commerce in Switzerland, whether made in Switzerland or in a foreign country, must carry the required checks, *contrôles*, and hallmarking *poinçonnées officiellement*.¹⁰⁷ (See Table 4.) The complexity of the dies and the engravings of the required markings with their minuscule size, multi-tiered motifs and fine detail may be viewed as security features.

The St. Bernard and the Branch Assay Mark together make up a single *poinçon officiel*, which is the Official Swiss Hallmark. The Common Control Mark is optional, but is carried on most watchcases. Nowadays, almost all marks are usually punched externally.¹⁰⁸ The Responsibility Mark may be individual, collective, or both. Karatage is optional, as it was long an industry custom.

The process of verification, inspection, assaying, and marking is not gratis. There are costs for the watchmaker, including the application for and registration of the Responsibility Mark and/or Collective Responsibility Marks, and identifiers. Each watchcase or bracelet (considered a separate article)¹⁰⁹ must carry the several marks, thus bearing the cost of assaying.^{110,111} When weighed against the price of a gold watch and the value of the assurance received, the cost (under about CHF10 for a watchcase with gold bracelet) seems trivial.¹¹²

When precious metal watchcases pass through the official testing channels, they are analyzed.¹¹³ In the past, the touchstone method was used;¹¹⁴ currently, X-ray fluorescence (XRF) is used because this method is more exacting.¹¹⁵ Not every watchcase is assayed,

105. See *Feuille Officielle Suisse du Commerce*, www.shab.ch; www.fosc.ch; www.fusc.ch

106. See Corti, Christopher, "Assaying Gold Jewellery-Choice of Techniques," *Gold Technology*, Issue No. 32, Summer 2001 (World Gold Council).

107. See OMCP Article 82.

108. Monney.

109. LCMP Article 7-2. The Bureau has the authority to make exceptions for technical reasons.

110. See OCMP Article 118. See also 941.319 "Ordonnance sur les Taxes du Contrôle de Métaux Précieux" (1998).

111. See L.F. 941.319 *Taxes*, Articles 7- 8; 11; 14.

112. See L.F. 941.319 *Taxes*, Section 2, Articles 6 through 14.

113. Compare, ICMP and OCMP 96-2. The law may be out-of-step with technology, but it states the touchstone method of assaying for watchcases, unless the results appear questionable and then the analytical assay is employed. The ICMP dated 1998 permits the use of the XRF in place of the touchstone. The analytical fire-assay is much more expensive at CHF 41 versus CHF6.60 for the touchstone test (941.319 "Ordonnance sur les Taxes"), but it is accurate to one part in 10,000. See Wälchli and Vuilleumier, "Touchstone Testing of Precious Metals," *Gold Technology*, No. 3, January 1991 at pp. 9-10 (World Gold Council).

114. See, Wälchli and Vuilleumier, Touchstone Testing of Precious Metals, *Gold Technology*, No. 3, January 1991, pp. 9-10 (World Gold Council). Because the touchstone test depends on a visual comparison of gold alloy color and the color variation among white gold alloys is negligible, the touchstone test is unsuitable for white gold alloys and contraindicated for platinum group metals.

115. Monney. Accord, Corti.

depending on the quantity and the different models submitted. In a lot of several watchcases of the same design, the Bureau checks a certain percentage by XRF and, a minimum of one randomly selected and complete piece (all the different parts composing the watchcase) for cupellation (fire-assay). To calculate the risks of selective sampling, it is important for the Bureau to know the provenance of the gold alloy, in other words, is the source or supplier of known reputation and/or certified by the Bureau,¹¹⁶ among other things.¹¹⁷

The age-old touchstone method is particularly suited to the testing of very valuable pieces, for which sampling by destructive means, such as scrapping, cutting, or drilling is unacceptable.¹¹⁸ Swiss assayers have a long tradition of successfully testing finished jewelry articles and have acquired, by experience, the ability to use the touchstone method at its maximum possible sensitivity. Thus, differences in precious metal content as small as 10 to 20 parts per thousand can often be established with confidence by the test.¹¹⁹ The modern X-ray fluorescence is also a non-destructive technique that is suitable for normal assaying requirements. It typically has an accuracy of 2-5 parts per thousand and is well-suited to the relatively flat and large surfaces of watchcases. It is a quick technique, taking about three minutes, and the results can be automatically printed out by the computer. It also measures the content of the other alloying metals present. It is not indicated for articles with chemical surface treatment or electroplating. The XRF instruments measure the intensity of the generated X-rays by wavelength dispersion analysis, but reference alloy standards of known composition close to that of the test piece are needed to ensure accuracy.¹²⁰ If the results of these tests are questionable, the more elaborate but totally destructive fire-assay, with an accuracy of one part in 10,000 is performed. The Swiss Precious Metals Control is accredited in accordance with SN ISO 17015 as a testing laboratory STS 071 for the analysis of gold, silver, platinum, and palladium.

The *Bureau Central du Contrôle des Métaux Précieux* protects consumers from fraud and manufacturers from unfair competition.¹²¹

116. The Bureau certifies *essayers du contrôle* and *essayers du commerce* who are charged with certain legal and ethical obligations. LCMP Articles 24 through 34; OCMP Chapitre V and VII.

117. Monney. Accord ICMP, the bureau makes periodic inspections of organizations dealing in precious metals. This allows the Bureau to verify that the regulations are met. Such inspections also build personal relationships and foster the exchange of information.

118. See Wälchli and Vuilleumier, p. 11.

119. See Wälchli and Vuilleumier, pp. 9-10.

120. See Corti.

121. Switzerland officially hallmarks more than one million articles per year. The Bureau uncovers about 2,000 disputes over fineness or forgeries in the same time. See generally, AFD website. In 2004, the Bureau punched 1,359,020 official hallmarks, along with 1,224,500 CCMs. The Bureau conducted 100 random checks of the industry and entertained 3,145 infractions/disputes. See *Forum D.*, May 2005.

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And so that is the significance of the marks on the back of your gold watch.

ACKNOWLEDGMENTS

I wish to thank several individuals who made my desire to compile and explain this information a reality. I especially wish to acknowledge the assistance of Daniel Monney, *Essayeur-juré* and Chief of Administrative Services of the Central Office for Precious Metals Control in Bern, for his patience and the wealth of information he provided, including the detailed information and illustrations of the Common Control Marks, Collective and Individual Responsibility Marks with identifiers, and information about the workings of the Bureau, branch offices, and precious metal watchcase industry, none of which could have otherwise been located. He also confirmed the technical accuracy of this essay regarding Swiss hall-marking and assaying. Dr. Lucien F. Trueb, a chemist, metallurgist materials scientist, and journalist, provided information concerning the gold watchcase sector. His work, *Gold in Watchmaking*, was a starting point for my inquiry. Edward Hahn, an engineer, photographer, and moderator of the TimeZone, provided the photo of the hallmarks on the Omega and encouraged me to continue my inquiry beyond "the 'J' is for Le Noirmont." Dr. Christopher W. Corti, of the World Gold Council, located important historical materials on assaying in Switzerland. Richard Madison provided the photo of the Jaeger-LeCoultre Responsibility Mark. John Davis of thePuristS.com provided the photos of the JLC Master Ultra-thin movement, crown, and dial.

ABOUT THE AUTHOR

Louis S. Flocco is a member of the Bar Association of Pennsylvania, New Jersey, and New York. He is a member of the NAWCC and Chapter 1, Philadelphia. Louis uses the pseudonym "Luger" on the online forums. His interests include the law governing the watch industry, French horological terms, and timekeeping apparatus whose movements are less than 50mm in diameter and 12mm in height.

Appendix 1.

Edelmetallkontrolle Contrôle des métaux précieux Controllo dei metalli preziosi

No 196 Montag, 13.10.2003 121. Jahrgang

Verantwortlichkeitsmarken für Edelmetall-, Mehrmetall- und Plaquewaren
Poinçons de maître pour ouvrages en métaux précieux, multimétaux et en plaque
Marchi d'arte per lavori di metalli preziosi, di pluri metalli e di placati

Zentralamt für Edelmetallkontrolle
 Bureau central du contrôle des métaux précieux
 Ufficio centr. per il controllo dei metalli preziosi

schweizerische – suisses – svizzeri

	Nr. 22296 Registrierungsdatum: 18. Juli 2003 Veri Haute joaillerie et objets d'art Fabrik und Detailhandel mit Uhren und Schmuck 3963 Grans-sur-Sierre
	No 22297 Date de l'enregistrement: 18 juillet 2003 Mélanie Meizoz Atelier de bijoux 1227 Les Acacias
	Nr. 22298 Registrierungsdatum: 21. Juli 2003 Manzoor Ahmad Dagg Gross- und Detailhandel mit Schmuck 4542 Luterbach
	Nr. 22299 Registrierungsdatum: 21. Juli 2003 Katalin Sipos Schmuckatelier 8700 Küsnacht ZH
	Nr. 22300 Registrierungsdatum: 22. Juli 2003 Jonas Princee Detailhandel mit Schmuck 8212 Neuhausen
	Nr. 22301 Registrierungsdatum: 22. Juli 2003 Michele Anderle Schmuckatelier 8803 Rüschlikon
	Nr. 22302 Registrierungsdatum: 23. Juli 2003 Andreas + Jane Fischer / Soleil D'Or Atelier und Grosshandel mit Goldschmuck 5504 Odemsingen
	No 22303 Date de l'enregistrement: 22 juillet 2003 Cogedis SA Commerce en gros de bijoux 1228 Plan-les-Orates
	Nr. 22304 Registrierungsdatum: 24. Juli 2003 Pirtili Kurt Stucki-Raatikainen Gross- und Detailhandel mit Uhren und Schmuck, Finnisches Kunsthandwerk 3011 Bern
	Nr. 22305 Registrierungsdatum: 30. Juli 2003 Bijoux Idee Tao-Tsan Detailhandel mit Schmuck 3011 Bern
	Nr. 22306 Data di registrazione: 13 agosto 2003 Orogen Sagl Atelier et commerce al dettaglio di gioielli 6900 Lugano
	Nr. 22307 Registrierungsdatum: 13. August 2003 Irène Meister / i.meister Schmuckatelier 5053 Münchenbuchsee
	No 22308 Date de l'enregistrement: 15 août 2003 Alexia Piaget Atelier de bijoux 2117 La Côte-aux-Fées
	Nr. 22309 Registrierungsdatum: 19. August 2003 Rahel Hidalgo Ganaha Schmuckatelier und Marktführer 9400 Rorschach

	Nr. 22310 Registrierungsdatum: 20. August 2003 Robert Günther Goldschmiede-Atelier 4053 Basel
	Nr. 22311 Registrierungsdatum: 21. August 2003 Simon Hitz Schmuckatelier 2502 Biel/Bienne
	No 22312 Date de l'enregistrement: 26 août 2003 Joseph Erard SA Fabrique de boîtes de montres 2340 Le Noirmont
	No 22313 Date de l'enregistrement: 2 septembre 2003 Dominique Vinciguerra Atelier de bijoux 2300 La Chaux-de-Fonds
	Nr. 22314 Registrierungsdatum: 4. September 2003 Illuck AG Grosshandel mit Schmuck der Marke Spectore 3014 Bern
	Nr. 22315 Registrierungsdatum: 4. September 2003 Rainbow-Shop Artemis, N. Zuber Detailhandel mit Schmuck 4142 Münchenstein
	Nr. 22316 Registrierungsdatum: 4. September 2003 Daniela Mühl Grosshandel mit Schmuck 5707 Seengen
	Nr. 22317 Registrierungsdatum: 8. September 2003 Felix Mehan Marktführer mit Schmuck 3013 Bern
	Nr. 22318 Registrierungsdatum: 9. September 2003 Gallet Group Inc. Atelier und Grosshandel mit Uhren und Schmuck 8702 Zollikon
	Nr. 22319 Registrierungsdatum: 12. September 2003 Jeannette Pfister / New feeling Detailhandel mit Schmuck 9000 St. Gallen
	Nr. 22320 Registrierungsdatum: 15. September 2003 Venturini Uhren-Schmuck-Juwelen, Nachfolgerin A. Elena Atelier und Detailhandel mit Uhren und Schmuck 9500 Wil SG
	No 22321 Date de l'enregistrement: 16 septembre 2003 Roula Sottile Commerce en gros de bijoux 1054 Bonnsens
	Nr. 22322 Data di registrazione: 16 settembre 2003 Zürcher & Co. SA Spedizioni internazionali 6830 Chiasso
	Nr. 22323 Registrierungsdatum: 19. September 2003 Veto-Schmuck Coppo & Morson Grosshandel mit Schmuck 6002 Luzern
	No 22324 Date de l'enregistrement: 19 septembre 2003 Stéphanie Harsch Atelier, bijoux 2520 La Neuveville
	No 22325 Date de l'enregistrement: 19 septembre 2003 Olivier Ernst Atelier, bijoux 1260 Nyon
	Nr. 22326 Registrierungsdatum: 23. September 2003 Manuela Paloncini Schmuckatelier 8050 Zürich

	No 22327 Date de l'enregistrement: 29 septembre 2003 Agnès Massera Atelier de bijoux 1890 St-Maurice
	Nr. 22328 Data di registrazione: 30 settembre 2003 Anrilia SA Atelier et commerce al dettaglio di gioielli 6900 Lugano
	No 22329 Date de l'enregistrement: 30 septembre 2003 Poiny International SA Commerce en gros de bijoux 1204 Genève
Erneuerungen – Rinnovi	
1278	Ulysse Nardin Le Locle SA, Manufacture et fabrication de montres et chronomètres, 3, rue du Jardin, 2400 Le Locle le 6 septembre 2003
1913	Richard Scherer, Goldschmied, Oberdorf 35, 9045 Trogen am 21. September 2003
1967	Peter Frischmuth-Frühling, Kellerswiesenstrasse 26, 9034 Eggenriet am 25. September 2003
4023	Kurt Züllig, Werkstatt für Schmuck, Linnausweg 9, 5408 Ennetbaden am 12. Februar 2003
4066	Walter Reitsch, Goldschmied, Turmweg 21, 3013 Bern am 16. Juni 2003
4087	Rolf Nopper, Gold-Silberschmiede-Atelier, Bünzlerfeldweg 2, 3998 Künz am 6. Juli 2003
4096	École Technique de la Vallée de Joux, Secrétariat et direction, 41, rue G.-H. Piguet, 1347 Le Sentier le 10 septembre 2003
4099	Bruno Schuch, Goldschmiede-Atelier, Thücherstrasse 27, 8156 Oberhasli am 24. September 2003
4103	Clinique du Bijou, Gérard Jacot-Descombes, 3, avenue Théodore-Flournoy, 1207 Genève le 29 octobre 2003
4104	Charles Veillon VPC SA, Vente par correspondance, 2, route de Bayre, 1030 Bussigny-Lausanne le 29 octobre 2003
Adress- und Firmenänderungen – Changements d'adresses et de raisons sociales – Cambiamenti dei indirizzi e delle ragioni sociali	
420	Piaget SA, 1, les Bâles-du-Temple, 2117 La Côte-aux-Fées nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
617	Vacheron & Constantin SA, Manufacture d'horlogerie, 1, rue des Moulins, 1204 Genève nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
1656	Piaget SA, 1, les Bâles-du-Temple, 2117 La Côte-aux-Fées nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
1709	Ossa Watch, Domenico Orsini, Goucherweg 4b, Postfach 205, 2557 Studen BE nouveau: Ossa Watch GmbH, Goucherweg 4b, Postfach 205, 2557 Studen BE
1765	Baume & Mercier SA, Mâtres Horlogers, 9, rue Le Royer, 1227 Les Acacias nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
2083	Piaget SA, 1, les Bâles-du-Temple, 2117 La Côte-aux-Fées nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
2566	Cartier Joaillerie SA, 46, rue du Rhône, 1204 Genève nouveau: Cartier SA, 11, rue du Marché, 1204 Genève
2567	Cartier Joaillerie SA, 46, rue du Rhône, 1204 Genève nouveau: Cartier SA, 11, rue du Marché, 1204 Genève
2920	Manufacture Usiflamm SA, Zone Industrielle de Moncor, 10, route des Biches, 1752 Villars-sur-Glâne nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
3496	Schilling AG, Goldschmiede & Schmuck-Manufaktur, Rheingold Schmuckmanufaktur AG, Vogesenstrasse 577, 4013 Basel nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
3748	Manufacture Jaeger-LeCoultre SA, La Colasse 8, 1347 Le Sentier nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
3938	Baume & Mercier SA, Mâtres Horlogers, 9, rue Le Royer, 1227 Les Acacias nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003

3997	Manufacture Jaeger-LeCoultre SA, La Colasse 8, 1347 Le Sentier nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
4066	Walter Reitsch, Goldschmied, Schlossstrasse 43, 3672 Oberlinsbach neu: Walter Reitsch, Goldschmied, Turmweg 21, 3013 Bern
4103	Gérald Jacot-Descombes, 3, avenue Théodore-Flournoy, 1207 Genève nouveau: Clinique du Bijou, Gérard Jacot-Descombes, 3, avenue Théodore-Flournoy, 1207 Genève
4533	Andreas + Karin Vögel, Goldschmiede, Schmuckdesign, Einkaufszentrum Tell, Tellstrasse 67, 5004 Aarau neu: Gloor Goldschmied AG, Rain 8, 5000 Aarau
4593	Kim Ong Teo, Postfach 1761, 8106 Adlikon b. Begenstorf neu: Kim Ong Teo, Badenerstrasse 362, 8004 Zürich
20223	Andreas Fischer, o/o Wolfgang Fischer-Leutenegger, Alte Hängingerstrasse 43, 5605 Dettikon neu: Andreas Fischer / OM-Inspiration Jewelry, Leuzburgerstrasse 9, 5504 Odemsingen
20229	Piaget SA, 1, les Bâles-du-Temple, 2117 La Côte-aux-Fées nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
20446	Vacheron & Constantin SA, Manufacture d'horlogerie, 1, rue des Moulins, 1204 Genève nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
20586	Pierre-Yves Turlo, 41, chemin William-Barbey, 1292 Chambésy nouveau: Pierre-Yves Turlo, 11, chemin de la Traversée, 1170 Aubonne
21010	Paul Emmenegger, Zur Goldschmiede, Dorfstrasse 6, 6205 Eich neu: Paul Emmenegger, Goldschmiedesteller, Stadtstrasse 44, 6204 Sempach Stadt
21095	Sogem SA, Société Générale de Montres, 12, avenue Industrielle, 1227 Carouge GE nouveau: Manufacture Roger Dubois SA, 2, rue André-de-Garriin, 1217 Meyrin
21144	IWC International Watch Co. AG, Baumgartenstrasse 15, 8201 Schaffhausen nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
21145	IWC International Watch Co. AG, Baumgartenstrasse 15, 8201 Schaffhausen nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
21201	Piaget SA, 1, les Bâles-du-Temple, 2117 La Côte-aux-Fées nouveau: Richemont International SA, 10, route des Biches, 1752 Villars-sur-Glâne le 29 octobre 2003
21223	Cartier Joaillerie SA, 46, rue du Rhône, 1204 Genève nouveau: Cartier SA, 11, rue du Marché, 1204 Genève
21226	Cartier Joaillerie SA, 46, rue du Rhône, 1204 Genève nouveau: Cartier SA, 11, rue du Marché, 1204 Genève
21454	Valérie Schaublin, 6, rue du Mont-Terri, 2800 Delémont nouveau: Valérie Di Meo / Versatil Bijoux, 28, rue de Chêne, 2800 Delémont
22260	Seraina Ebner, Alte Stockstrasse 41, 5022 Rombach neu: Seraina Houriet, Goldschmiede, Alte Stockstrasse 41, 5022 Rombach
Löschungen – Radiazioni – Cancellazioni	
1968	Gabor SA, Alte Lyssstrasse 12, 3270 Arberg
1970	Roland Fritsche, Hauptstrasse 102, 9400 Rorschach
4084	Jean Lassale SA, 10, rue de Blagnac, 1227 Carouge GE
4092	Norbert Eschmann, 30, chemin du Piquet, 1025 St-Gulpie VD
4101	Martin Fellmann, Friedheimstrasse 8, 8057 Zürich
4102	Walter Jucker, Goldschmiede Son Miggel, Stradung 11, 7460 Savognin
4105	Interkornal SA, o/o Rosemarie Leonard, 28, chemin Auguste-Vilbert, 1218 Grand-Saconnex
4106	Michel Schmid, 5, chemin des Pralles, 1092 Belmont-sur-Lausanne
4107	Marian von Sawilski, Brigglicker 18, 8050 Zürich
4108	Gérald Rigolet, 14, rue Louis-de-Savoie, 1110 Morges
4109	Fredin SA, 24, chemin des Jonchères, 2022 Bevaix
4111	René Lea, Via al Cioso 11, 6929 Gravesano
4112	Viviane Zäch, Obegartenstrasse 3, 8705 Erlenbach ZH
4113	Pierre Mommier, Bijouterie, Villa Huchette, 1342 Le Pont
4117	Fox Interiors, Barandun & Sturzenegger, Torquasse 6, 8001 Zürich
4120	Alice Müller-Oster (od. Xuster), Breilacherstrasse 11, 8702 Zollikon
4121	Speidel Imports AG / El Sol AG, Venedigstrasse 37, 4053 Basel
4667	Schilling AG, Goldschmiede & Schmuck-Manufaktur, Vogesenstrasse 577, 4056 Basel
Kollektiv-Verantwortlichkeitsmarke Poinçons de maître collectif Marchi collettivi d'arte	
No 1/78 Quoetz SA Fabrique de boîtes de montres 2350 Städelrieder	

Appendix 2.

An excerpt from the official database of the Swiss customs. Dated 11/29/2001.
This is a sample listing, please refer to the NAWCC Library and Research Center
for a complete listing of case maker tables.
Translations are in parentheses below.

Poinçon de maître collectif no 1 (marteau sans manche) (Hammer without handle)

Affichage d'une seule marque		Retour à la recherche		Affichage plus marques	
No	1	☐ Seul les marques non radiées?			
Titulaire Owner					
☒ PM ☒ MIF ☐ Essayeur ☐ Fondeur		Nom: Union Suisse pour l'Habillage de la Montre (USH) Adresse: 6, rue du Comouiller Case postale 4026 NPA / Lieu: 2500 Biel/Bienne 4 Canton: BE Pays:			
(Registered) Enregistrement		(Renewal) Echéance		(Expiration) Radiation	
11.07.1934		10.03.2006			
Remarque					
Nur für Uhrgehäuse aus Gold und Platin, Minimaldicke 0,15 mm Seulement pour boîtes de montres en or et platine, épaisseur minimum 0,15 mm Herkunft: Kanton Neuenburg und Jura / Origine: cantons de Neuchâtel et Jura (Only for watchcases in gold or platinum; minimum thickness .15mm.)					
Date mutal:		No séquence		Auteur mutal:	
29.11.2001		9989		DMONEY	
(Origin: Neuchatel or Jura Regions)					

No	Nom de l'utilisateur (User's Name)	Lieu (City)	(Registration)	(Expiration)
			Enregistrement	Radiation
2	F. Baumgartner SA	Genève	21.07.1934	05.11.1958
11	Manufacture Taubert	Genève	21.07.1934	05.11.1958
26	Ponti, Gennari & Cie.	Genève	21.07.1934	05.11.1958
100	Union Suisse pour l'Habillage de la Montre (USH)	Biel/Bienne 4	11.07.1969	
101	Junod Frères & Cie.	La Chaux-de-Fonds	21.07.1934	12.09.1935
102	Amez-Droz & Cie. Boîtes de montres Zéma, Jean Bor	Tramelan	21.07.1934	31.08.1986
103	Pro-Boîtes SA	La Chaux-de-Fonds	21.07.1934	07.02.1995
104	Manufacture de Haute Horlogerie Cartier S.A.	La Chaux-de-Fonds	21.07.1934	28.02.2000
105	Les Fils de Jules Blum	La Chaux-de-Fonds	21.07.1934	18.05.1974
106	Louis Wursten fils	La Chaux-de-Fonds	21.07.1934	01.01.1941
107	Mme H. Cattin, Successeur de François Cattin	La Chaux-de-Fonds	21.07.1934	13.03.1952
108	Jacques Beiner / Roger Parel SA	St-Imier	21.07.1934	05.01.1984
109	Oréada, Manufacture de boîtes SA	La Chaux-de-Fonds	21.07.1934	
112	G. & C. Ducommun SA	La Chaux-de-Fonds	21.07.1934	01.01.1940
115	Manufacture Favre & Perret SA	La Chaux-de-Fonds	21.07.1934	
116	Flückiger & Cie	La Chaux-de-Fonds	21.07.1934	13.03.1952
117	Guillod Gunther SA	La Chaux-de-Fonds	21.07.1934	
119	Gindraux SA	La Chaux-de-Fonds	21.07.1934	30.08.1972